Starquill: Gameplay Loop & Systems Overview

Audience: product manager / commissioner **Date:** 2026-07-02 · reflects the build as implemented (~460 automated test methods; last full confirmed run green) plus locked-in design decisions. Anything not yet playable is explicitly marked.

The pitch

Starquill is a mobile idle RPG for Android. A party of four hand-drawn "paper doll" characters battles endless waves of monsters on its own; the player's job is the fun part — firing abilities at the right moment, deciding which loot to wear, and choosing when to take on quests and dungeons. Every piece of equipment is visible on the character wearing it, so getting stronger always *looks* like something. The game respects the player's absence: the party keeps earning while the app is closed.

The core loop (every 5–30 seconds)

1. **The party fights by itself.** Four characters auto-attack the current enemy wave once per second.
2. **The player fires Verbs.** Three ability cards (e.g. Fireball, Slash, Heal) rotate through a shared hand; tapping one unleashes it. Every ability and every enemy has one of six classic stats (STR, DEX, CON, INT, WIS, CHA) arranged in two rock-paper-scissors triangles — Physical: STR beats DEX beats CON; Mental: INT beats WIS beats CHA. Matching the right ability to the right enemy is the core skill.
3. **Kills pay out.** Gold always; equipment ~15% of the time (with "pity" guarantees so rare items can never dry up forever: an Uncommon within 50 kills, up to a Legendary within 5,000). Rarely, a **dungeon key** or a **fragment** drops alongside.
4. **Loot changes the doll.** Equipping an item immediately redraws it on the character — hat, armor, weapon, all of it.

The session loop (every few minutes)

- A **travel bar** under the gold counter fills as waves are cleared. Quests can be discovered randomly at any wave (3%), and the bar *guarantees* one on arrival (every ~20 waves) — the player is never starved of the next goal.
- A discovery pops a **banner**; tapping it shows the quest: name, difficulty tier, wave count, reward preview, and a line of zone flavor dialogue. Accept or decline.
- **Quests are the structured challenge.** Each zone is a ladder of 11 quests — Normals, two Elites (mini-boss), two Hards (mixed enemy types), then the **Zone Boss**. Enemies are themed to the zone (Verdant Hollow: brawny beasts; Gloomspire Ruins: arcane horrors). During a quest, the travel bar becomes a wave-progress bar.

- **Dungeons are the targeted hunt.** Spending a key opens a timed wave rush (2½–3½ minutes) where *every* drop matches the key's promise — a Brown Helm key showers brown helmets. Clearing waves past par earns bonus reward rolls, so Verb skill directly buys loot. Mini-boss every 5th wave; the bell is the only ending — no fail state, and quitting early (even closing the app) banks what was cleared.
- **No fail state anywhere.** Quests can be retreated from anytime, keeping half the gold, and retried free — the quest waits.
- **Completion pays curated rewards:** gold bonuses and guaranteed-quality items (Elites guarantee Uncommon+, Hards and Bosses Rare+; dungeon floors scale with key difficulty up to guaranteed Legendary), revealed on reward sheets. Elite/Hard/Boss quests also pay keys. Beating the zone boss unlocks the next zone.
- Completing quests is what raises **quest level** — the single dial that scales enemy strength, gold value, and loot quality everywhere. Quests are the engine of progression; exploring is the flywheel between them; dungeons are where the player spends agency on exactly the loot they want.

The meta loop (across days)

- **Equipment depth.** Every item carries a primary + secondary stat and most carry an **awakened ability** — a passive bonus that levels up over time while worn, or instantly for gold. Rarity governs everything: stat budget, ability ceiling, sell value.
- **Party building.** The player owns a roster of characters across species, fields four, and swaps them to cover stat matchups. Characters gain XP from kills and quests and level up, granting stat points weighted toward their species strengths; benched roster members earn a half share so nobody stagnates.
- **Training.** Each character has a per-level gold Training ladder (+5% damage per rank, exponentially priced) — the always-available gold sink that keeps player power growing exponentially alongside enemy difficulty. Catch-up pricing keeps new recruits affordable.
- **Key crafting.** Keys come in three kinds — **color** (loot's literal color, 8 huntable families), **type** (equipment slot), and **class** (one of 15 two-stat build archetypes). Fusing keys for gold combines their promises and stacks their difficulty: a triple-fused "Brown Warrior Helm key" is the late-game precision hunt, priced quadratically. Off-kind keys sell for gold — no dead drops.
- **Offline earnings.** The party keeps exploring while away at 50% efficiency, capped at 8 hours; a "Welcome back" claim sheet presents the haul on return — doubled by watching an ad.
- **The shop** sells timed boosts (auto-firing Verbs, faster ability rotation, priced as minutes-of-income), a free 4-hour chest (ad-doubleable), and the **Tavern**: six generated characters on a 6-hour rotation, shown bare-headed so their faces read, reviewable head-to-toe before spending gold to recruit. Recruits step straight into an open party slot.
- **Monetization is rewarded-ad-first and never pay-to-win** *(ships in MVP)*: watch an ad to double offline earnings or the timed chest; a one-time Remove Ads purchase. Interstitials appear only at quest completion and are removed by the purchase. If a reward arrives while the bag is full, it waits in a mailbox — never lost.

How the systems interlock

Combat produces gold, loot, and keys → loot raises stats (and its abilities grow while worn) → stronger parties clear quests and deeper dungeon rushes → quest level rises → everything scales up, demanding better loot and smarter stat coverage → which quests provide broadly and keyed dungeons provide precisely. Gold is the pressure valve with four sinks: ability level-ups, Training ranks, shop boosts, and key fusion — with item- and key-selling (single tap, or "Optimize All" to auto-equip the whole roster) recycling the inventory's 50 slots.

Where the game is headed (designed, not yet built)

- **Locations** (Destinations second half) — the fragments counter's purpose: procedurally generated one-time places (*Blackroot Castle*, *Bogun's Bog*) discovered while exploring, visited as short boosted runs that end in a **dialogue choice** — recruit the stranger, take the treasure, start a quest, or fight the boss. Design of record: `docs/plans/2026-07-02-destinations-design.md` §4 (Sprints D2-D3).
- **Prestige** — a costly reset for permanent multipliers (Stellar Ink). All economy formulas already carry the multiplier, stubbed at 1.0.
- **Verb collection** — abilities currently come with the character; the design adds Verb drops and a collection layer.

Current status & honest gaps

Playable now	Explore combat, Verbs, full loot/equip/sell loop, paper-doll rendering, party management + Training, complete quest loop (discover → accept → fight → complete/retreat → boss → next zone), keys + fusion + dungeon rushes, shop boosts, 4h chest, offline earnings claim, character leveling, reward mailbox, save/load
Sprint 12 (next)	Tutorial + unlock gating, animation/juice, final balance pass (destinations knobs untuned), Full UI Pass 2, real ad/IAP store wiring, bug sweep, Android build, ship
Known gaps	Art is placeholder-flat outside the character/paper-doll work (Full UI Pass 2). Ad/IAP run against test/mock backends until store accounts are wired at ship. Destinations economy knobs are design-doc guesses pending the sim branch.